

ADVERSARIAL ML & RAG EVALUATION

Document Fraud Detection Proxy & Hallucination-Free Knowledge Retrieval Engine

1. Project Objective

This project implements a complete, reproducible machine learning pipeline to evaluate the robustness of AI-based document fraud detection systems against adversarial attacks. Beyond measuring model accuracy, it benchmarks defense strategies, analyzes decision boundary behavior, tracks experiments through MLflow, and provides an evidence-grounded Retrieval-Augmented Generation (RAG) assistant for stakeholders to query experimental results using natural language.

2. Solution Overview

The implementation consists of six integrated engineering components:

Component	Purpose
Data Pipeline	3-class document fraud proxy (Genuine, Tampered, Forged) with deterministic seeding.
FraudCNN	Custom PyTorch CNN trained from scratch for clean baseline accuracy.
Adversarial Evaluation	Red-team attack benchmarks via FGSM, PGD-20/40, and C&W; (IBM ART).
Defense Benchmarking	Benchmarks Adversarial Training (PGD) vs. Feature Squeezing.
Explainability	Decision boundary & feature space visualizations (PCA, UMAP, DeepFool).
Executive RAG Assistant	Offline AI chatbot for natural language Q&A; over MLflow experiment logs.

3. Key Results

Metric	Result
Baseline Clean Accuracy	88.82%
Validation Accuracy	88.88%
Clean Test Loss	0.2806
PGD-40 Accuracy (Baseline)	25.00%
Best Robust Accuracy	37.50%
MLflow Experiments Tracked	73 Runs
Automated Test Status	31/31 Passed

4. Engineering Highlights

- **Modular Architecture:** Decoupled package hierarchy separating data, models, attacks, defenses, and RAG layers.
- **End-to-End Reproducibility:** Fixed deterministic execution across PyTorch, NumPy, and CUDA.
- **Natural Language Analytics:** Evidence-grounded RAG assistant preventing LLM hallucination during executive inquiries.

5. Deliverables

Deliverable	Status
GitHub Repository & Documentation	Verified
FraudCNN Registered Model	Verified
MLflow Experiment Tracking	Verified
Attack Evaluation Report	Verified
Defense Comparison Report	Verified
Decision Boundary Visualizations	Verified
MLflow Export Package	Verified
RAG CLI Assistant	Verified
Gradio Executive Dashboard	Verified
Benchmark Evaluation Dataset	Verified

8. Release Status

The implementation satisfies all mandatory project objectives. Core functionality has been experimentally validated, documented, regression-tested, and packaged under a **Zero-Regression release policy**.